

Duration: 3 Hours**Maximum Marks:35**

Q1 Create a DataFrame containing information about different products with attributes such as Product_name, Price, Rating, and Number_of_Sales. Generate a Dendrogram to identify groups of similar products. Display a suitable title and labels for the Dendrogram.

[10 Marks]

Q2. Write a Python program to create a DataFrame with the attributes Temperature, Humidity, Wind Speed, Pressure, and Rainfall. Apply the following data transformation techniques and display the transformed data:

- a)Min-MaxScaling
- b)Standardization
- c) Normalization

[20 Marks]

OR

Q2. Implement a K-Nearest Neighbors (KNN) classifier on the given Student Grading dataset to classify students into different performance categories. Perform the necessary data preprocessing, train and test the model, and evaluate its performance using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

[20 Marks]

Q3. Viva

[5 Marks]